Factor bases, solving degree and arity in elliptic-curve index calculus: what is explored, what is not

Date. 2026-10-24. **Kind.** Desk research over the four repositories (`crypto` at `d51b908a77`, branch `feat/koblitz-norm-base-20261023`; `cryptanalysis`; `crypto-autoresearcher`; today's `ic-regime-review-20261024/REVIEW.md`) plus a three-lane literature sweep (factor-base constructions; solving-degree reduction; large-arity decomposition). Nothing new was run. Every repository number is quoted from the file named beside it; numbers marked *derived here* are this note's arithmetic on quoted numbers and are not measurements. No security claim about any deployed curve is made or withdrawn.

The question asked. Across the factor-base strategies that lower the solving degree (degree of regularity) or the factor-base size, there seem to be two trade-off curves: solving degree against factor-base dimension, and cost against the number of summands $m = 3 \dots 9$. What other strategies exist, which have never been tried, and which tried ones were not exhausted?

---

0. Bottom line

1. **The two curves are real, and the repository has measured one half of each.** Solving degree against factor-base dimension ℓ is measured on binary curves at $m = 3, 4$ only (chained S_3 presentation), and on prime-field PKM towers at $m = 2, 3, 4$ with $N = m \cdot t \le 20$. Cost against arity is *derived* at $n = 131$ for every m (the free-oracle floor and the per-target budget), but **no algebraic solving degree has been measured at $m \ge 5$ in any of the repositories**. The literature has exactly three $m \ge 5$ data sets, all at $\ell \le 3$ and $n \le 29$: Semaev 2015 (chained S_3, $n \le 16$, F4 degree 4 in 2,300 systems), Huang–Petit–Shinohara–Takagi 2015 (the same chain, $n = 13$–29, $m = 5, 6$, measured D_{reg} **4.3–5.0**, 13–34 hours and 44–102 GB per instance), and Karabina 2015 (two S_4 links with one auxiliary point, $m = 5$, $n \le 19$, $D = 7$–8). Nobody has $m \ge 5$ with $\ell \ge 4$. The $m \ge 5$ half of the arity curve is the largest unexplored region in both the program and the field, and the first cells are reachable with the repository's existing closure instrument.
2. **The shape of a single subspace is closed as a lever on the degree; presentation and position-dependence are not.** On binary curves at prime n, every F_2-subspace shape sits at random once the trace class is conditioned on (the one measured effect was an artefact), and the only closure a subspace can have is bounded by linear Kneser and already attained by geometric progressions (`RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md`). Yield is conserved for every set of a given size (`AR:KN-FIND-007`). What moves the degree is how the same ideal is presented to the solver: fall-closure/F4 against XL (2–3 degrees), the 2-torsion quotient $E/\langle T \rangle$ (bit-degree 7 ⟶ 5 per unknown), the symmetric `e_k ∈ V^{(k)}` formulation (unknowns Σ `dim V^{(k)}`, 15–40× cheaper on progressions), auxiliary variables (PKM towers on prime fields, the chain on binary), and, in the literature only, drawing each summand from a different coset or block (Galbraith–Gebregiyorgis, Diem): the one base-side change measured to lower the F4 degree, never built here.
3. **The deciding measurements are known, specified, and mostly unrun.** In order of information per unit cost: (a) a **Δ-sweep at fixed (m, ℓ)**: every published "degree 4 flat to $\ell = 8$" cell sits at overdetermination $\Delta = n - m\ell$ of 13–76, the repository's rise past 4 at $\ell = 7$ sits at $\Delta = -2, 0$, and no one has held (m, ℓ) fixed while moving Δ; (b) the **Galbraith–Gebregiyorgis $m = 4$ formulation** (binary Edwards t-coordinate, $(Z/4)^{m-1}$ ⍰ S_m invariants), the published best $m = 4$ Gröbner result ($\ell = 4$ in 9–52 s at $n \le 53$), one to two orders cheaper than the repository's chain at the same (m, ℓ), and closed in the repository on a different experiment; (c) the first **closure degrees at $m = 5, 6$** on the chain at $N \le 56$, which fit the current engine; (d) the FGHR-symmetrised $m = 4$ system, whose unknown count $4\ell - 3$ is independent of n (the chain's is $2n + 4\ell$), never built because `symmetrised_terms` returns `None` for $m \ge 4$; (e) any $m = 4$, $\ell \ge 4$ cell at $\Delta \ge 0$ (every measured one is at $\Delta \le -3$); (f) the guess count that holds the closure at degree 4 past $\ell = 7$ (one cell, 8 draws); (g) a fitted elimination exponent w on the real Macaulay matrices, since the $m = 4$ price moves by about 40 bits between $w = 1$ and $w = 2.3$. Items (d)–(g) are on the 2026-10-18 card `backlog/XP-20261018-m4-solving-degree-equivariant/`; (a)–(c) are new here.
4. **A derived what-if shows why the arity curve matters.** Under the one bounded-degree observation the repository has (fall closure at degree 4, held to $N = 54$), a formulation whose unknown count does not grow with n would put $m = 5$–6 within roughly 4–15 bits of rho at $n = 131$, where the chained presentation stays 16–47 bits above it across $m = 4$–8 (§1.3, derived here, heavily caveated). Whether the degree stays bounded on such a formulation is exactly what no one has measured, and the published heuristics (Petit–Quisquater's $D \approx m^2 + 1$ first fall degree; Semaev's bounded-Macaulay-degree assumption) disagree about it by the whole exponent.
5. **For prime fields the map is closed to one open line and one closed loophole.** The PKM tower is the only presentation with sub-$|F|$ membership degree, and its F4 degree grows about one degree per eight unknowns at $m = 4$ ($D \ge 8$ at $N = 20$), with $m = 3$ dying at 99.6 GiB inside degree 8. The autoresearcher's approved, unbuilt "non-enumerative membership oracle" (`EXP-ICEX-153c34`) is, by its own coordinator's Chinese-remainder argument, enumeration in a changed basis: `F_p[x]/(g_V) ≅ F_p^V`. Only auxiliary variables escape that, which is why PKM is the only line left.

---

1. The two curves, as measured

1.1 Solving degree against factor-base dimension ℓ (binary, F_2-subspace bases)

System: Weil descent of the chained S_3 system for m summands in an ℓ-dimensional random subspace $V \subset F_{2^n}$, $N = (m - 2) \cdot n + m \cdot \ell$ Boolean unknowns, first fall degree 3 everywhere (Kosters–Yeo trace equation). Three instruments read three different degrees on the same draws (`RESEARCH_DREG_MEASUREMENT.md` Results 4–14; `ic_next10_20261014/06-equivariant-f4-m4/README.md`):

m	ℓ	plain XL refutation degree	msolve F4 degree	MutantXL fall-closure degree	cells, surplus
3	2	5	4	3–4	N = 11–18, S = −1…+6
3	3	5–6	4	4	S = −5…0
3	4	6–7	4	4	S = −5…+1
3	5	7 exactly at (10,5); ≥7 at (11,5) (13,5) (15,5)	4 at (10,5), (15,5)	4	S = −5…0
3	6	not run	not run	4 (8/8, N = 34, 36)	S = −4, −2
3	7	not run	not run	> 4 (8/8, N = 40, 42); degree-5 closure capped at 169 GB; one guessed variable restores 4 on 6/8 draws	S = −2, 0
4	1	5	3–4	3–4	
4	2	6	3–4	3–4	S = −3…0
4	3	≥7	4	4	S = −3, −2
4	4	not measured	4 (4/4 at (11,4)); (13,4) capped at 3 GB	4 (N = 42)	S = −5, −3
4	5	not measured	capped	4 (8/8 at (11,5) (12,5))	S = −9, −8
4	6	not measured	not run	4 (2/2 at (15,6), N = 54, 25.8 GB)	S = −9

Readings the repository itself draws:

- **Which degree is which.** The closure column is the quantity every published experiment reports (Magma's F4 step degree) and, on refuted instances, equals the last fall degree exactly (Caminata–Gorla 2023, §5.2). The XL column has no published counterpart.
- **The XL fits $\lceil \ell/2 \rceil + 4$ (m = 3) and $\ell + 4$ (m = 4) are XL's, not the ideal's.** The closure and F4 read 4 flat across $m \in \{3, 4\}$, $\ell \le 6$, $N \le 54$. The rise at $m = 3$, $\ell = 7$ is the first bounded-degree failure; whether it is one degree or many is unmeasured (Result 12–14; dreg_closure_m4l6_20261018/RESULTS.md refutes "$m \cdot \ell \ge 21$" and leaves "$\ell = 7$" as the surviving explanation).
- **The degree is set by (m, ℓ) and is flat in n at fixed ℓ** (Result 10). The semi-regular reference, which counts unknowns, overstates growth in n.
- **Every m = 4, ℓ ≥ 4 cell is at surplus S ≤ −3** (underdetermined), chosen because unsatisfiable draws were affordable there. Index calculus runs at $S \approx 0$. No m = 4 cell at $S \ge 0$ with $\ell \ge 4$ exists (XP card §3, "Not measured anywhere").
- **One structural number predicts the m = 2 degree exactly** (cryptanalysis experiments/pdp-degree-heuristics/README.md, 656 bases, 7 families, 9 fields): with e = n − dim V − dim V^{(2)}, the closure degree is D = 2 + [e < −3.5] + [e < −8.5] (MAE 0.06). Geometric progressions minimise dim V^{(2)} = $2\ell - 1$ and are 15–40× cheaper per query where a random base needs one more degree. At m = 3 the direct formulation's degree is set by system size, the family moves it by at most one degree; in the **symmetric formulation** (unknowns are coordinates of e_k ∈ V^{(k)}) the unknown count *is* Σ_k dim V^{(k)} (15 against 19 at $\ell = 3$; 21 against 34 at $\ell = 4$) and progressions are 500–1,500× cheaper where the random base needs one more degree. This is the one place where factor-base shape still moves cost, and it is a presentation effect (the shape sets the unknown count of the symmetric system), not a degree effect.
- **Solver cost slopes per unit of ℓ** (cryptanalysis experiments/pdp-scaling/README.md, Koblitz n = 17…61, planted instances, fits $\log_2 s = a + c \cdot \ell + d \cdot n$): meet-in-the-middle c = 1.93 (m = 3), 1.89 (m = 4), 2.48 (m = 5, range too short for the 2^{3ℓ} half); WDSat 3.02; WDSat with the trace clause 2.51; CryptoMiniSat 3.95 (m = 3), 6.97 (m = 4); msolve 5.25 (m = 3). Every algebraic or SAT engine has a steeper slope than enumeration; the trace-clause variant is the only one within 0.6 bits of enumeration's slope.

1.2 Solving degree against tower depth (prime fields, PKM towers)

System: S_{m+1} plus t quadratic tower equations per summand, N = m·t auxiliary unknowns, |V| = 2^t (docs/ic/factor-base-encyclopedia/README.md §0.1; pkm_m3_n21_msolve_20261018; REVIEW §1 item 3):

m	N = m·t	F4 degree D	status
2	2…18	4, 4, 4, 4, 4, 5, 5, 5, 5	plateau at 5
2	20, 22	6, 6	rises
3	9, 12, 15, 18	6, 6, 7, 7	
3	21	≥ 8, run died at 99.6 GiB inside degree 8	formally inconclusive; the review puts the prior on linear growth
4	8, 12, 16	6, 7, 7	
4	20	≥ 8 after three productive degree-8 rounds, 5 GB cap	plateau broken: about one degree per eight unknowns

D does not depend on the prime, on $v_2(p - 1)$ up to 96 (P-224), or on Kummer against isogeny against Dickson-torus tower (82 cells, PR #1942). A deployed size needs $t \approx 56$–64 at m = 4 with admissible degree about 3; measured is 7–8 at $t \le 5$.

1.3 Cost against arity m at n = 131 (ECC2K-130), derived

Repository model (RESEARCH_ECC2K130_DECOMPOSITION.md §5; RESEARCH_ECC2K130_DESCENT_DEGREE.md §2): relations 2^{ℓ}, targets per relation $m! \cdot 2^{n - (m-1)\ell}$, linear algebra $m \cdot 2^{2\ell}$, oracle C(N, ≤D)^w per target. Rho reference 2^60.81 with the $\langle -1 \rangle \times \langle \pi \rangle$ discount. All n = 131 rows are extrapolations.

m	ℓ at the floor	free-oracle floor	per-target oracle budget (ops)	search space per target $2^{(m-1)\ell}$	required speed-up over search
2	43.3	2^89.25	2^−27.86	2^43.3	2^−71.2
3	33.0	2^68.58	2^−6.78	2^65.0	2^−71.8
4	26.8	2^56.40	**2^+5.71** (≈ 52 operations)	2^77.9	2^−72.2
5	22.8	2^48.44	2^+13.94	2^86.5	2^−72.5
6	19.9	2^42.85	2^+19.77	2^92.5	2^−72.8
8	16.1	2^35.61	2^+27.35	2^100.5	2^−73.2

Three facts about this curve that the repository states and that frame everything below:

- **For any enumerative or table oracle the curve is flat.** The product `relations × targets × oracle ≈ m·2^n` is independent of ℓ, and the required speed-up over the oracle's own candidate set is `≈ 2^−72` at every `m`. Corrigan-Gibbs–Kogan (`S·T² ≥ r/|Aut|`) and the 2025 Dinur–Keller–Marmor non-adaptive bound close every table or k-sum design regardless of `m`.
- **The budget per target rises steeply with `m`** (52 operations at `m = 4`, `2^14` at `m = 5`, `2^20` at `m = 6`, `2^27` at `m = 8`), while the yield falls as `2^{−(m−1)ℓ}`. So the arity curve for an *algebraic* oracle is a race between a budget that grows about `2^{7(m−4)}` and a system whose size and degree grow with `m`.
- **The presentation decides which way the system grows.** Two unknown counts at the floor's ℓ, *derived here*:

`m`	ℓ	chained S_3: `N = (m−2)·131 + mℓ`	FGHR-symmetrised direct `S_{m+1}`: `N_B = m(ℓ−1) + 1`	degree-4 columns `C(N, ≤4)` chained / FGHR	per-target budget (ops)
3	33	230	97	`2^26.8` / `2^21.8`	`2^−6.8`
4	27	370	105	`2^29.5` / `2^22.3`	`2^5.7`
5	23	508	111	`2^31.4` / `2^22.6`	`2^13.9`
6	20	644	115	`2^32.7` / `2^22.8`	`2^19.8`
8	16	914	121	`2^34.8` / `2^23.1`	`2^27.4`

The chained unknown count is dominated by `(m − 2)·131` intermediate-point coordinates and grows with `m`; the FGHR count is independent of `n` and nearly flat in `m`. **A what-if, derived here and not a measurement.** Hold the fall-closure degree at 4 (measured to hold on the *chained* system only, to `N = 54`, `m ≤ 4`), take the repository's measured closure exponent `w ≈ 2.2` (`dreg_closure_driver_20261016`) and its ≈ 8-bit column-touch to rho-step conversion (REVIEW §1 item 3). Then the per-target oracle costs `2^51`–`2^69` rho steps on the chained system and a nearly flat `2^40`–`2^43` on the FGHR system, against budgets that rise about 7 bits per unit of `m`. Excess over rho, in bits:

`m`	chained, degree 4 held	FGHR direct, degree 4 held
4	+47	+31
5	+35	+15
6	+26	+4
8	+16	nominally below rho

The chained `m = 4` row agrees with the review's independent +37 to +49 bracket at degree 5. Three caveats bound the rest of the table. (i) **It assumes a bounded degree at `m ≥ 5`, which nothing has measured**: the repository's degree data stop at `m = 4`. (ii) On the FGHR direct system the Boolean degree of the generators themselves grows with `m` (the plain-x descent has degree at most `Σ wt(e_i)`: 6, 12, 15–20 at `m = 3, 4, 5`; cryptanalysis `pdp-scaling`), so a degree-4 Macaulay space cannot contain them at `m ≥ 5` without auxiliary variables, and the FGHR gain measured at `m = 3` is mostly the change of curve to `E/⟨T⟩` plus one cover variable (`lit2026_w8_summation_models_20261011`); the FGHR rows at `m ≥ 5` are therefore illustrative only. (iii) On the chained system the generator degree is fixed at 3 for every `m`, so its rows have no such obstruction: the only unmeasured quantity is the closure degree as `N` grows past 54. The reading that survives the caveats is this: **under any bounded-degree assumption the arity curve improves with `m`, because the budget grows faster than a bounded-degree column count; so the first closure-degree measurement at `m = 5` is the single most informative unmeasured point on the curve, and on the chained system it is well posed and affordable** (`m = 5, ℓ = 3, n = 13` has `N = 54`, the same size as the `(15, 6)` cell that ran in 25.8 GB).

What the literature says the optimum `m` is. Three heuristics and one theorem, none of which reaches `n = 131` (lane report `lit_large_arity.md` §A5, §(b)):

source	assumption	optimal `m` at given `n`	claimed crossover with rho	status
Petit–Quisquater 2012	`D_reg ≈ D_ff ≤ m² + 1` (first-fall-degree assumption)	`m = 2` for `n ≤ 200`, 3 at 500, 4 at 1,000–2,000, 9 at 20,000; asymptotically `m ≈ n^{1/3}`, `ℓ ≈ n^{2/3}`	`n ≥ 2,000`	assumption refuted experimentally (Kosters–Yeo, Huang–Kosters–Yeo, Kousidis–Wiemers: `D_reg` rises with `n` at fixed `m`; S_5 Weil descent reaches `D_reg = 13 = m(m−1)+1` at `m = 4`)
Semaev 2015	chained S_3, F4 degree `≤ 4` for every `m`	`m ≈ √(2 ln 2 · n / ln n)`: 6 at `n = 100`, 7 at 150, 8 at 200, 12 at 500	`n > 310` (so ECC2K-130 is **not** reachable even under his assumption)	assumption contradicted at `(n, m) = (45, 2)` and `(25, 3)` (`D_reg = 5`); HPST 2015 measure 4.3–5 at `m = 5, 6`
Karabina 2015	`D_reg ≈ 9` at `m = 5`	`m = 5`, `ℓ = n/5`	`n ≈ 1,200`	extrapolation from `n ≤ 19`
Diem 2013 (rigorous)	none	`m = min(⌈√log₂ q⌉, ⌊n/2⌋)` with each summand in its own direct-summand subspace	subexponential `L(3/4)` for `a·log^{1/3} q ≤ n ≤ b·log q`	gives nothing for `F_{2^n}`, `n` prime
Galbraith–Gebregiyorgis 2014 (practical)	measured	"the best we can seem to hope for in practice is `m = 4` and `ℓ ≤ 10`"	none: `2^{n−40} ≥ 2^{n/2}` trials for `n > 100`	measured to `m = 4, ℓ = 4` (Gröbner, 9–52 s at `n ≤ 53`) and `ℓ ≤ 7` (SAT)

Joux–Vitse state the program's exact question as an open problem (JoC 2013 §6): with a subspace of dimension `d > 1` and `m < n`, "the best values for `d` and `m` involve a trade-off that depends heavily on the cost of the decomposition test", and the equations lose the `S_{md}` symmetry, keeping only `S_m`; their invariant-ring attempts "have been unsuccessful so far". No published work has resolved it.

1.4 The third axis: surplus (overdetermination)

`RESEARCH_DESCENT_CROSSOVER.md` proves the identity that the overdetermination `S = n − mℓ` of the descended system and the log of the expected decomposition count are the same parameter: one bit of yield per surplus equation. So "pick ℓ a little below `n/m`" (Shantz–Teske's δ; Joux–Vitse's `(k−1)`-point decompositions on extension fields) buys a cheaper refutation at a yield cost that is exact. The repository's surplus control (Result 6) shows the degree rising only at `S ≤ −4`; at `S ≥ −2` every `m = 3` cell reads 6 (XL) / 4 (closure). The DoR campaign (`factor_`

base_dor_campaign_20261013, PR #1826) measured δ at m = 3: shrinking ℓ from 3 to 2 at n = 15 drops XL D* from 5 to 3. At m = 4 the positive-surplus side of this axis is unmeasured for ℓ ≥ 3.

---

2. Factor-base strategies: inventory and verdicts

The full inventory is docs/ic/factor-base-encyclopedia/README.md (A1–A13 prime, B1–B2 extension, C1–C11 binary, D1–D10 Koblitz, E1–E6 Jacobians). Compressed by what each family was *for*:

aim	family	verdict	bounding result
lower membership degree (binary)	F_2-subspace, any shape (coordinate, random, progression, Frobenius progression, trace-zero, Möbius pole, scaled, inversion-closed)	**closed at m = 2**: trace class is a free bit; closure dim V·V ≥ 2ℓ − 1 by linear Kneser, attained by progressions; at m = 3 shape moves D by 0 at ℓ ≤ 4 (G1, G3, G5 killed)	Hou–Leung–Xiang; RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md §2.2–2.3
	subfield F_{2^ℓ}	one degree below random, **does not exist at prime n**; on b = 1 the yield collapses (points lie in a subgroup)	
Frobenius-stable set at prime n	invariant subspaces	**empty**: dimensions {0, 1, 130, 131} at n = 131	$\mathrm{ord}_{131}(2) = 130$
	Hamming weight in a normal basis	rejected: calls grow as |F|^{1.9–2.5}; SAT at n = 23 decides nothing in 180 s	RESEARCH_HAMMING_IDEAL_PDP.md; pdp_followups_20261010
	cyclotomic γ-weight C_w	yield-neutral (p = 0.36 against the null), 2–12% costlier to build; **solver clause untested** (one n = 11 cell went the wrong way)	cyclotomic_falsifier_20261015
	quasi-subfield polynomials	barren at n = 131; β > 1/2	RESEARCH_QUASI_SUBFIELD.md
	quadratic-form sets Tr(c x^{2^e+1}) = 0 (G4)	**proposed, never built**	shape-search §7
raise yield at equal D	τ-twisted ordered chain $R = P_1 + \tau P_2 + \tau^2 P_3$ (G2)	**survives at n ≤ 9** (degree rise 0); ×m! yield unread; gate G6 (n = 31) never run	shape_search_g1_g6_20261015
	any set geometry (prime)	**theorem**: mean yield depends only on |F|; coverage headroom ≤ 1.58	AR:KN-FIND-007
fewer columns (Koblitz)	signed Frobenius orbits; Z[τ] small-norm multipliers; integer multiples; CM multipliers	constants (|Aut|, multiplier set size); table route only; "does not help the algebraic route" (asserted, not measured)	ic_improve_20261023; CGK bound
sub-|F| membership degree (prime)	PKM towers (Kummer, isogeny, Dickson torus)	**the only open prime-field line**; D grows ≈ 1 per 8 unknowns at m = 4; needs 256–512 GB for the next cells	encyclopedia §0; REVIEW §1.3
	one-binomial cosets, Dickson sets, intervals, orbits, Eisenstein, bit-box, Möbius images, isogeny-augmented	refuted or constant: behave as random sets of their size	yield conservation; pair-table law S = 0.75·r^{1/6}; Bezout B ≤ 3d
	small-root (Coppersmith) membership	closed: reach δ = 1/80 against 1/5 needed at m = 4	RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md
	non-enumerative Gröbner membership (EXP-ICEX-153c34)	approved, unbuilt; coordinator's own prior 0.95 negative: F_p[x]/(g_V) ≅ F_p^V (CRT), so it is enumeration in a changed basis	AR:H-ICEX-6ca21f
intermediate field	Gaudry–Diem x ∈ F_q; Joux–Vitse (k−1)-points; FGHR Y-line; D_4-symmetrised S_5 at k = 4	the only measured sub-rho exponents live here (and in the genus-3 cover); constants 18–850×; stacked D_4 + 2LP never run	encyclopedia §3

Results that bound every family at once: yield conservation; sparsity invariance (every generating set of the membership ideal of a fixed set has the same Hilbert function, so re-encoding without auxiliary variables cannot lower D*); the pair-table family law; the symmetry inventory (every fold is a constant or polynomial in n); Bezout; subgroup confinement; "index calculus is priced by the ambient field, rho by the subgroup"; and the band-priced frontier (no IC arm beats rho at any weight on any measured curve).

---

3. Explored but not exhausted

Ranked by information per unit cost. Each row names the gap, the evidence that it is a gap, and the cheapest next cell.

1. **FGHR-symmetrised m = 4 system (arm B)**
   - *Done:* FGHR 2-torsion frame built and measured at m = 2, 3 (koblitz_symmetrised.rs; 30–416× on extension fields; E/⟨T⟩ identity certified)
   - *Gap:* symmetrised_terms returns None for m ≥ 4; every m = 4 degree is on the chained system with 2n intermediate unknowns. The "m = 4 is one degree above m = 3" reading may be the chain's, not the arity's
   - *Next:* build S_5 in $(w_1..w_4, w_R, s)$, exhaustive gate, measure closure degree at ℓ = 2…5, n ≤ 19
   - *Cost:* 1–2 days code, hours of compute (XP card M1)
2. **m = 4 at surplus S ≥ 0, ℓ ≥ 4 (arm A)**
   - *Done:* all m = 4, ℓ ≥ 4 cells at S ≤ −3
   - *Gap:* the regime index calculus runs in is unmeasured at m = 4
   - *Next:* (16, 4), (20, 5) on the chained system, closure at degree 4
   - *Cost:* runnable today; N ≤ 60, tens of GB
3. **Guess count that holds degree 4 past ℓ = 7**

- *Done:* Result 14: one fixed variable restores degree 4 on 6/8 draws at `m = 3, ℓ = 7`; both branches of one guess refute on 5/8
- *Gap:* one cell, one ℓ; the hybrid-closure cost curve `2^g · cols(4)^w` against ℓ is what prices the `ℓ ≥ 7` regime, and the whole `m = 4` price swings on it
- *Next:* `g` against ℓ at `m = 3, ℓ = 7, 8` and `m = 4, ℓ = 6, 7` with the dense closure
- *Cost:* 25–70 GB per cell

4. **Fitted elimination exponent `w`**
   - *Done:* `w` assumed at 1 and 2; measured ratios `log₂ ops / log₂ cols = 2.18–2.33` at one size (`dreg_closure_driver`); `K` measured by Callgrind
   - *Gap:* the `m = 4` price is `+10` bits at `w = 1`, `+46` at `w = 2`, `+37…+49` in rho units across `w ∈ [2, 2.33]` (REVIEW F1)
   - *Next:* record rank profile and density per degree block of the real matrices; try structured Gaussian / block Wiedemann for `w < 2` (REVIEW D1)
   - *Cost:* days, no new memory
5. **`C₃ ⊂ S₄` Faugère–Svartz blocks**
   - *Done:* G2 (Frobenius) equivariant pass built, bit-identical, 2–3× against `gf2_elim`; useless in production because random `V` is not σ-stable
   - *Gap:* the summand-permutation symmetry applies to every system; `S₄ ⊃ C₃` is odd-order, characters in `F₄`; specified, unbuilt
   - *Next:* after #1: two blocks (trivial over `F₂`, conjugate pair over `F₄`), each ≈ ⅓ dimension
   - *Cost:* 2–3 days (XP card M2)
6. **G2 τ-twisted ordered chain**
   - *Done:* survives at `(5,2), (7,3), (9,3)` with rise 0; yield rises for random bases
   - *Gap:* `×m!` never read because fields saturate; G6 at `n = 31` never run
   - *Next:* `n = 13…19` with the real F4, read yield and degree together
   - *Cost:* hours
7. **Cyclotomic base solver clause**
   - *Done:* yield-neutral; build cost +2–12%
   - *Gap:* the only value it could have (lower solving degree of the membership system) was never tested above `n = 11`
   - *Next:* F4 on `S₃` + cyclotomic membership at `n = 23`, `w = 2, 3`, against ONB Hamming base
   - *Cost:* hours (single-word plug-in limits `n`)
8. **G4 Frobenius-stable quadratic-form sets**
   - *Done:* proposed 2026-10-01
   - *Gap:* never built; the only other Frobenius-stable family with an algebraic membership test at prime `n`
   - *Next:* yield census via `ic search`; F4 on `S₃` + `2k` quadratic constraints for `D`
   - *Cost:* 1–2 days
9. **Multiplier closure on the algebraic route**
   - *Done:* `Z[τ]`, `Z[φ]`, integer multiples: 2–20× on the table pipeline; "does not help the algebraic route" asserted
   - *Gap:* unmeasured. Norm-2 multiplier `τ̄` *is* the 2-torsion symmetrisation (`x ↦ x + 1/x`); higher norms with non-rational kernels add a degree-`N(α)` rational constraint per summand, not a symmetry
   - *Next:* one toy cell: `m = 3`, `ℓ = 3`, base `V ∪ τ²(V)` with the degree-4 `x`-map constraint; compare degree and yield
   - *Cost:* hours; expected negative
10. **Closure degree at `m = 5, 6`**

- *Done:* `S₆` (190,252 monomials) and `m = 5, ℓ = 4` descents exist in cryptanalysis (`pdp-scaling`, SAT and MITM only); the rotated-`m5/m6` admission note (2026-09-25) demands a canonical corpus and a `>64`-variable engine for `n = 131`-shaped systems
- *Gap:* **no algebraic degree at `m ≥ 5` in any repository**; the literature has three toy sets at `ℓ ≤ 3` (Semaev, HPST15, Karabina; §5.1) and none at `ℓ ≥ 4`. The first cells need no engine work: the chained system at `(m, ℓ, n) = (5, 2, 11)`, `(5, 3, 13)`, `(6, 2, 11)` has `N = 43, 54, 56` unknowns, under the `u64` cap, and the same size as the `(15, 6)` cell that ran at 25.8 GB
- *Next:* `dreg_closure` on those three cells, four unsatisfiable draws each (`chained_solution_count` already generalises in `m`)
- *Cost:* hours to a day on a 32 GB host; the multiword engine (XP card M3) only for `ℓ ≥ 4`

11. **Last fall degree (Huang–Kosters–Yang–Yeo) and Crossbred on the degree cells**

- *Done:* Crossbred tried only on the `m = 2` toy oracle at `n = 7, 9`; LFD never instrumented
- *Gap:* order-independent degree instruments (REVIEW D3)
- *Next:* run on the committed 92 draws
- *Cost:* hours

12. **Memory-light degree bound via GF(2) block Wiedemann on the Macaulay matrix**

- *Done:* block Wiedemann exists over `Z/rZ` only
- *Gap:* would give the plain-XL degree (2–3 degrees pessimistic) for cells past `(4, 6)` without renting memory (REVIEW D2)
- *Next:* port `koblitz_sparse_la.rs` to GF(2) consistency test
- *Cost:* days

13. **Stacked `D₄ + 2LP` at `k = 4`**

- *Done:* `D₄` 18.2× constant; 2LP lowers the LA exponent 0.495 ⟶ 0.420 but costs 2.31× whole-solve; sparse LA null blamed on the completion rule
- *Gap:* the only unrun combination in the one regime with a measured sub-rho exponent
- *Next:* ecbench end to end with `sparse-cado` and a full-rank stop rule (REVIEW M3)
- *Cost:* one week

14. **PKM `m = 3` at `N = 21`**

- *Done:* died at 99.6 GiB inside degree 8; `m = 4` plateau already broken
- *Gap:* formally inconclusive; the review's cheap check: finish target 1 unspecialised to the end of degree 7 and count solutions (F3)
- *Next:* one msolve run on a ≤ 128 GB host
- *Cost:* hours

15. **Published-IC reference arm**

- *Done:* every "× faster" is against rho or the program's own producer; the in-house F4 is 10^3–10^7 operations off a brute grid on PKM systems (`generator_effect_ic_20261019/E3.md`)
- *Gap:* no FGHR+msolve+WDSat baseline at matched `(q, n, m)` exists in the repos
- *Next:* protocol F2 of the review
- *Cost:* 2–3 days to specify, a week to run

16. **Galbraith–Gebregiyorgis `m = 4` formulation (closed on the wrong experiment)**

- *Done:* the encyclopedia's C8 verdict "GG buys variables, not degree" rests on GG14's *special subspace* at `m = 3` in a symmetric-S_4 encoding at `n = 9, 15` (`factor_base_dor_campaign_20261013` F6). The 4-torsion was pursued only as the `π − 1` transport on K_0 (`RESEARCH_EXOTIC_COORDINATES.md` §10–11, negative: a quarter the yield at equal cost)
- *Gap:* GG14's **Experiment 2** is a different object: binary Edwards model, coordinate `t = x + y`, factor base `t ∈ V`, invariants under `(Z/4)^{m−1} ⨝ S_m` (`e₁, s_j = Π(t_i² + t_i)`), Magma F4, **`m = 4, ℓ = 4` in 9–52 s at `n = 17…53`, degree 4**, `ℓ ≤ 7` with SAT. The repository's best `m = 4, ℓ = 4` is msolve on the chain at `(11, 4)`: 386–805 s, and `(13, 4)` capped at 3 GB. The published formulation is one to two orders cheaper at the same `(m, ℓ)` and larger `n`, and ECC2K-130's $E(\mathbb{F}_2) \cong Z/4$ is exactly the torsion it needs
- *Next:* build GG14 Exp. 2 natively (binary Edwards form of K_0, `t`-subspace base, 4-torsion invariants), gate against enumeration, measure closure degree and cost beside the chain on the same `(n, ℓ)`
- *Cost:* 2–3 days

17. **Symmetrised (e_k) presentation above the `ℓ = 6` crossover**

- *Done:* `RESEARCH_DEGREE_REDUCTION.md` L2: symmetrisation is the wider presentation below `ℓ = 6`; the instrument reached `ℓ ≤ 3`; "needs a solver this repo does not have"; withdrawn
- *Gap:* HPST 2013 measured exactly that crossover from the other side: at `n = 53` their symmetric-plus-nonsymmetric variable system (≈ `(m²ℓ + mn)/2` variables, `D_reg` 3–4 against FPPR's 6–7) is **12× faster at `ℓ = 5` and 85× at `ℓ = 6`** (6.68 s vs 80.8 s; 34.8 s vs 2,967 s). GG14 Exp. 2 reaches `ℓ = 8` at `m = 3` in 52 s. cryptanalysis has the `e_k ∈ V^{(k)}` formulation at `m = 2, 3` only
- *Next:* run the `e_k` formulation (and the HPST mixed one) at `m = 3`, `ℓ = 6…8`, `n = 29…53` with msolve, beside the chain
- *Cost:* days

18. **Chain topology and auxiliary count**

- *Done:* the binary builder is a linear chain (`m − 2` free intermediates); a `tree` mode exists only in the prime-field PKM code and was never run there either ("`m = 5` and `tree` were not run", `RESEARCH_PKM_TOWER_ORACLE.md`)
- *Gap:* HPST 2015 report the balanced split $\{S_3(x_1,x_2,w_1),\ S_3(w_1,X,w_2),\ S_3(w_2,x_3,x_4)\}$ "better balanced" than the linear chain at `m = 4`; Karabina's one-auxiliary split ($S_4 \cdot S_4$ at `m = 5`, $5\ell + n$ unknowns against the chain's $5\ell + 3n$) trades unknowns for degree (`D = 7`–8). The axis "number of auxiliary points `k ∈ {0 … m − 2}`" has only its two endpoints measured anywhere
- *Next:* add `tree` and `k`-auxiliary modes to `build_decomposition_system`; measure closure degree against `k` at `m = 4, 5`, toy `n`
- *Cost:* 1 day

19. **Symmetry-broken (disjoint) factor bases at large `m`**

- *Done:* G2 (τ-twisted ordered chain) is the Frobenius special case; survives at `n ≤ 9`; `×m!` never read
- *Gap:* GG14 Exp. 3 (affine translates `V + v_i`): `D_reg` often 3 instead of 4, gain `(m − 1)!`; Gebregiyorgis thesis §3.7 (disjoint block subspaces): success probability `2^{mℓ}/2^n` with **no `m!`**; Diem 2013 (direct-sum decomposition, rigorous); APS 2018 (partitioned base); PKM tower code has a `disjoint` symmetry mode. The `m!` in the program's targets-per-relation term is `2^9.5` at `m = 6` and `2^15.3` at `m = 8`, against `m×` more relations and `m²×` linear algebra (`2^5.2` at `m = 6`), so the lever grows with `m` while the linear algebra it costs shrinks in share
- *Next:* add a `disjoint` mode to the binary builder; read yield and degree together at `m = 4, 5`
- *Cost:* 1 day

20. **Δ-sweep at fixed `(m, ℓ)`**

- *Done:* surplus control Result 6 at `m = 3`: degree rises at `S ≤ −4`, flat 6 (XL) / 4 (closure) at `S ∈ [−2, +1]`; the `ℓ = 7` rise is at `S = −2, 0`
- *Gap:* every published "degree 4 flat to `ℓ = 8`" cell is at `Δ = 13`–76 (§5.2); no one has the same `(m, ℓ)` across Δ. If the closure degree is a function of Δ, the ℓ-laws are artefacts of the `Δ ≈ 0` sampling
- *Next:* `m = 3, ℓ = 7` at `n = 21, 25, 29, 37` (Δ = 0 … 16) with the closure; same at `m = 4, ℓ = 5`
- *Cost:* hours per cell at `N ≤ 64` (`n ≤ 29` at `m = 3`); the engine port for larger `n`

21. **Trace relation added explicitly at degree 1**

- *Done:* the Kosters–Yeo fall is observed at degree 3 on every draw; the trace-zero base pre-filter never fires on odd-order targets
- *Gap:* whether part of XL's growth is re-deriving a relation that is linear on solutions (lane §5.2); one flag in `build_decomposition_system`
- *Next:* re-run the committed 92 draws with the trace row appended
- *Cost:* hours

4. Not explored at all

From the repository analysis (literature-derived additions follow in §5):

- **The m ≥ 5 half of the arity curve for algebraic oracles**, in any formulation (§1.3). The budget per target at m = 5–8 is 2^14–2^27 operations, 8–22 bits more generous than at m = 4; the repositories have nothing there, and the literature has three toy data sets at ℓ ≤ 3 (§5.1), none at the ℓ ≥ 4 where the trade-off lives.
- **A formulation whose unknown count is independent of n at m ≥ 4.** The FGHR direct system has N_B = m(ℓ − 1) + 1; the FPPR/WDSat symmetric system has Σ_{k ≤ m} dim V^{(k)} (≈ m(m+1)ℓ/2 on a progression, also n-independent). Both avoid the (m − 2)·n intermediate coordinates that make every measured m = 4 cell cost what it does. The symmetric formulation exists in cryptanalysis at m = 2, 3 (pdp-degree-heuristics §6) and nowhere at m ≥ 4.
- **The surplus axis at m = 4, ℓ ≥ 3, S > 0** (the Shantz–Teske / Joux–Vitse lever on binary curves at the arity that matters).
- **A degree instrument that is not the dense closure** (block Wiedemann over GF(2); last fall degree), so that cells past (4, 6) can be bounded without a 256 GB host.
- **The N1 group-algebra lift** (F_2[Z/263] redundant representation; exact-cancellation census at n = 23) and the **G4 quadratic-form sets**: both have one-afternoon falsifiers written in docs/ic/RESEARCH_IC_NOVEL_DIRECTIONS_20261007.md and RESEARCH_FACTOR_BASE_SHAPE_SEARCH.md §7 and neither has been run.
- **Position-dependent bases in any form** (affine cosets, coordinate blocks, Diem's mixed dimensions): the only shape lever the literature measured to *lower* the degree (GG14, degree 3 against 4 at large n), and the only one whose value grows with m (§3 row 19, §5.3). The repository's G2 is its Frobenius special case and was never read for yield.
- **Diem's exact-surplus device**: δ = dm − n summands from a subspace one dimension smaller, so that the condition count equals n exactly. The repository tunes surplus only in steps of ℓ; at m = 4, ℓ = 33, n = 131 the chain has S = −1, and Diem's device would make it 0 at no yield cost beyond the exact one bit.
- **The Galbraith–Gebregiyorgis m = 4 formulation itself** (§3 row 16): the repository has the ingredients (2-torsion frame, Z/4 on K_0) but never the binary-Edwards t-coordinate system with (Z/4)^{m−1} ⧄ S_m invariants, which is what made m = 4, ℓ = 4 cost seconds in 2014.

Things that look unexplored but are closed by a theorem or an identity, so that they are not re-proposed:

- Any cleverer *set* of the same size for a table oracle (yield conservation; pair-table law; CGK; DKM non-adaptive bound).
- Any re-encoding of a fixed set's membership without auxiliary variables (sparsity invariance, Proposition 1 of RESEARCH_INDEX_CALCULUS_FACTOR_BASE.md).
- Moving along the isogeny class to lower the degree (a_6 enters below the leading form at every m; d_reg is constant on the class, proved).
- Subspace shape at prime n (trace class + Kneser).
- Gröbner membership over F_p for an explicit set V (CRT: it is enumeration in a changed basis). This covers the Amadori–Pintore–Sala random-point base too (§5.3).
- Two-level bases with intermediates restricted to a larger subspace W: the yield cost per intermediate is 2^{dim W − n}, which the surplus identity makes exact; it cannot pay (*derived here*).
- F4 trace reuse (Joux–Vitse F4Remake) for Boolean systems: success probability ≈ 0.29^{steps} over F_2 by the authors' own analysis (§5.2).
- Frobenius-equivariant blocks on a single decomposition system: the ideal is not Frobenius-stable for a random target (§5.2).

---

5. What the literature adds

Three lane reports sit beside this note: lit_factor_bases.md, lit_degree_reduction.md, lit_large_arity.md. Each has a data table with URLs. What they add to the repository's own record:

5.1 Large arity and the m–ℓ trade-off (lit_large_arity.md)

- **Largest summation polynomial ever computed:** FHJRV (Eurocrypt 2014) P_{ϕ,8} in characteristic 2 with one 2-torsion point, 470,369 monomials, 40.5 CPU-hours by sparse interpolation of a resultant of P_6 and P_4; P_7 has 2,247 terms (383 s), P_6 has 50 (plain S_6: 638 in the elementary symmetric variables; plain S_7, S_8 never finished). The per-variable degree is still 2^{m−1}; the gain is sparsity and an m-fold smaller base. The only decomposition experiment with it is m = 5 (P_{ϕ,6}) over F_{(2^31)^5} at ℓ = 1: 10.28 s per test. **No m ≥ 6 decomposition exists anywhere.** The "S_9" claim in one 2019 paper is unsourced.
- **m ≥ 5 on binary curves:** Semaev 2015 (n ≤ 16, F4 degree 4), HPST 2015 (n = 13–29, ℓ = 2, 3, D_reg 4.33–5.00, 13–34 h, 44–102 GB; "D_reg slightly larger than 4 when m = 5, 6, so FFDA must probably be relaxed"), Karabina 2015 (m = 5, n ≤ 19, D = 7–8 with one auxiliary point). HPST15's rise from 4.0 at ℓ = 2 to 4.3–5.0 at ℓ = 3 is the only published hint of a D(ℓ) dependence at m ≥ 5; the repository's closure instrument has never been pointed at it.
- **Plain Weil descent of S_{m+1} without symmetrisation is hopeless in m:** Kousidis–Wiemers measure D_ff = D_reg = 7 = m(m−1)+1 at m = 3 and **13** at m = 4. That is why every large-m formulation chains or symmetrises.
- **Shantz–Teske 2013:** at m = 2, shrinking ℓ below n/2 ("Delta method", Δ ≈ 2⌊(n−15)/6⌋) pays more than the 2^{m−1} yield it costs; m = 2 beats m = 3 until n ≈ 290 (linear algebra) and in relation gathering until n > 600; the hybrid method gives less than Delta and does not stack with it. This is the repository's surplus identity (§1.4) measured from the other side.
- **Generic m-sum:** only meet-in-the-middle applies (T = N^{⌈m/2⌉}, M = N^{⌊m/2⌋}); Wagner, Schroeppel–Shamir, Howgrave-Graham–Joux, dissection and the Nikolić–Sasaki/Dinur trade-offs all need a quotient or an order, which a prime-order curve group lacks. No generic-group lower bound for k-sum is published; the program's CGK and Dinur–Keller–Marmor arguments are the sharpest available. With MITM the product law becomes m!·2^{n − (m−2−⌈m/2⌉)ℓ}: flat for m ≤ 5, improving by ℓ bits per two units of m beyond that at memory 2^{⌊m/2⌋ℓ}, still 2^{114} at m = 8, ℓ = 16 on ECC2K-130.
- **"Split completely" transplants exist and are closed:** quasi-subfield polynomials (Huang–Kosters–Petit–Yeo–Yun 2020) are the elliptic-curve version of the function-field-sieve trick; Euler–Petit and McGuire–Sheekey rule out linearized families (β ≥ 3/4 against β ≤ 0.1 needed). No Bluher-polynomial transplant to point decomposition was found.
- **Courtois 2016** claims a memory-free 2^{n/3} oracle for m = 2, ℓ = n/2 by guessing a small multiple K·R and a common divisor; unverified, and the repository measured only his linearised half (2^{91} at n = 131).

5.2 Solving-degree reduction (`lit_degree_reduction.md`)

- **Vocabulary mapping that reframes the repository's data.** Every experimental paper (Petit–Quisquater 2012, Shantz–Teske 2013, HPST 2013/2015, Galbraith–Gebregiyorgis 2014, Semaev 2015, Karabina 2015, Kousidis–Wiemers 2019) reports Magma's F4 *step degree*, which re-multiplies degree falls: it is the repository's **MutantXL closure degree**, not its plain-XL degree, and it equals Caminata–Gorla's solving degree `sd` (JSC 2023, Def. 1.1). Proved: `sd = max{d_lf, max GB degree}`, so on a refuted instance with field equations **the closure degree is exactly the last fall degree**; `sd ≤ d_reg + 1` (Salizzoni 2023); plain XL `≤ 2·d_reg − 2` (Semaev–Tenti 2021). Consequences: (i) the repository's XL laws `⌈ℓ/2⌉ + 4` and `ℓ + 4` are **new**: nobody has published a plain-XL refutation degree on these systems; (ii) its closure data (4 flat to `m·ℓ ≤ 24`) agree with the literature's "`D_reg = 4`" at `m = 3…6`; (iii) Kousidis–Wiemers' `D_ff = 7` (`m = 3`) and `13` (`m = 4`) are for the plain Weil descent of `S_{m+1}`, whose generators have degree `m(m−1)+1`; the repository's "first fall degree 3 everywhere" is for the chain, whose generators have degree 3. Not a conflict; worth writing up as such.
- **Every published "flat at 4" sits at large overdetermination.** GG14: `n = 37–97`, `ℓ ≤ 8`, so `Δ = n − mℓ = 13–76`; HPST13: `n ≤ 53`, `mℓ ≤ 18`; Semaev's own remark: the degree exceeds 4 as soon as the system is under-determined. The repository's rise past 4 at `m = 3, ℓ = 7` is at `Δ = −2, 0`, a regime no paper sampled; its surplus control (Result 6) ran `Δ ∈ [−5, +1]` only. **A Δ-sweep at fixed `(m, ℓ)` is the cheapest discriminating experiment**: if the degree is a function of Δ rather than of ℓ, the `⌈ℓ/2⌉ + 4` law is an artefact of `Δ ≈ 0`, and the cost model's yield identity (one bit per surplus equation) prices the trade exactly. Gaggero–Gorla's threshold (`n > C(N, D)` equations makes degree `D` generic) and Semaev–Tenti's "sufficiently overdetermined" theorem give the curve to compare against.
- **The only rigorous growth statement is linear in ℓ and n-free.** Derived by the lane (not in any paper; hypotheses to check): encode `x_i ∈ V` upstairs with the linearised polynomial `L_V` and apply Huang–Kosters–Yang–Yeo 2018 Thm 1.1 to `{S_{m+1}, L_V(x_1), …, L_V(x_m)}`; the last fall degree of the descended system is at most about `2m²·ℓ + O(m)`, independent of `n`. Loose (≈ 90 against measured 4–7 at `m = 3, ℓ = 5`) but it fixes the *shape*: flat in `n` is plausible and partly proved; **flat in ℓ has no supporting theorem and contrary data at `m = 2`** (step degree 3 for `ℓ ≤ 8`, 4 for `9–18`, 5 for `19–24`, Kosters–Yeo and Kousidis–Wiemers, up to 208 GB). No lower bound exists for Weil-descent systems; the only one in the area is Yokoyama et al. 2020 for the naive prime-field method.
- **Closed by the literature, so not worth building:** F4Remake / trace reuse (G1 of the novel-directions note) has success probability ≈ `0.29^{steps}` over F_2 (Joux–Vitse's own `c(q)` analysis), measured gains of 2–3.4× only over `F_p`; Faugère–Svartz blocks cannot act on a single decomposition system because the Weil-descent ideal is Frobenius-stable only when `V` is stable *and* `x_R ∈ F₂`, which fails for every random target (the $C_3 \subset S_4$ summand-permutation block of §3 row 5 is unaffected).
- **Constant-factor items not in the repository:** adding the Kosters–Yeo trace relation explicitly at degree 1 to the XL matrix (does part of the `⌈ℓ/2⌉ + 4` growth come from XL re-deriving a relation that is linear on solutions?); Macaulay-submatrix "degree smoothing" (Jaques–Ran–Samardjiska–Seitner, ePrint 2026/408), a fractional-degree XL worth a few bits where the `C(N, D)` jump between consecutive degrees is large; a guess count calibrated against the Gaggero–Gorla threshold `n > C(N − g, D)` instead of the degenerate hybrid sweep.
- **Nothing from 2022–2026 attacks these systems directly.** Adjacent: Caminata–Gorla 2023, Salizzoni 2023, Gaggero–Gorla 2023/24 (theory), Vidal–Ionica–Delaplace 2024 and ePrint 2026/360 (Crossbred), ePrint 2024/1923 (a prime-field index-calculus implementation).

5.3 Factor-base constructions (`lit_factor_bases.md`)

28 constructions tabulated with URLs; those not in the repository's inventory, and what each is worth here:

- **Position-dependent (disjoint) bases**: GG14 affine cosets `F_i = V + v_i` (measured, `m = 3`, `ℓ = 5–7`, `n = 37–97`, degree often 3 against 4); Gebregiyorgis coordinate blocks (proposed); Nagao's "disjoint factor base" (ePrint 2015/984, claims success probability ≈ 1 under the first-fall assumption); **Diem II's mixed-dimension pair** `U ⊃ U′` with `dim U′ = dim U − 1` and exactly `δ = dm − n` summands from `U′`, which is the device behind his rigorous `e^{O(log(q^n)^{3/4})}` and is the only published construction that tunes the surplus to zero *exactly* rather than in steps of ℓ. §3 row 19 and §1.4. Accounting note (*derived here*): at equal column count a disjoint design has lower yield than one subspace of the same size by about `e^m/√(2πm)`; its gain is at equal *system size per summand*, where it removes the `m!` for `m×` columns, so it favours large `m` where linear algebra is cheap.
- **Amadori–Pintore–Sala "dynamic" base (prime fields)**: the base is the random points `R_i = u_iP + v_iQ` themselves, one relation from `S_m` with each `X_i` constrained by `Π_j(X_i − x(R_j))`, no linear algebra; the only prime-field construction with end-to-end DLP timings (22-bit `p`, `m = 3, 4, 5`: 268–388 s). **Closed by the repository's own argument**: `F_p[X]/(Π(X − x_j))` is CRT-isomorphic to functions on the grid, so the Gröbner computation is enumeration of `s^m ≈ p` grid points in a changed basis (encyclopedia §7 item 2, `AR:H-ICEX-6ca21f`); Yokoyama et al. 2020 prove the matching lower bound for this "naive" structure. APS's crossover claims are against PKM, not against rho.
- **Frobenius-stable bases on subfield curves** (Galbraith–Granger–Merz–Petit, SAC 2020): one decomposition system per Frobenius orbit, n^2 on linear algebra, Frobenius as a second symmetry-breaking source. This is the repository's signed-Frobenius-orbit and barrel formulation, re-derived without the paper (`AR:DEC-20260916-3c0cf5`); its full text, the most useful blocked document in the sweep, would show which of GGMP's several Frobenius-stable bases they found cheapest. **Tsakou–Ionica 2021** generalise to any endomorphism φ with `φ^k = ±1`: factor base `{P : Tr_φ(x(P)) ∈ F_q}` or the norm, imposed as `n − 1` linear conditions in a normal basis while keeping `S_{m+1}` in `x`. Extension-field and GLS/GLV curves only; theoretical for elliptic curves. The binary-prime-`n` analogue is the repository's `Z[τ]`-multiplier closure, which the lane confirms is Koblitz-only.
- **Gorla–Massierer trace-zero hyperplane section** (`n = 3, 5`): a *geometric* base (the `F_q`-points of `V_n ∩ {x_0 = … = x_{n−3} = 0}` in Weil-restriction coordinates) for the DLP in the trace-zero subgroup `T_n`; a full 36-bit `T_3` DLP in 957 s. Not on the list, but it attacks a different group (`T_n`, not `E(F_p)` or `E(F_{2^n})`), and the repository's A10 experiments show why trace-zero sets do not transfer to the prime-field target: relations land in the wrong subgroup.
- **Kosters–Yeo's measured base preference**: at `m = 2` the "`deg < k`" window kept degree 4 at `n = 40` where random subspaces grew. The repository's shape search explains most such coordinate-versus-random gaps as the trace class (§2) and the residual as Kneser closure, so this is covered.
- **Sieving**: no sieve exists for the summation-polynomial decomposition on an elliptic curve itself; every published sieve (Joux–Vitse 2012, Sarkar–Singh 2014/15, Vitse–Wallet 2015) acts on a Jacobian through a one-parameter family of functions and a univariate splitting test. The only route to a sieve is a cover to a Jacobian, which the repository has (E2, E6).
- **Largest measured instances on record**, for calibration of the reference arm (§3 row 15): WDSat `m = 3` at `(n, ℓ) = (89, 11)`: 85,834 s satisfiable, 175,718 s refuted, under 20 MB; Gröbner on the same model stops at `ℓ = 7` (35 GB), `ℓ = 8` exceeds 200 GB; `m = 2` at `(41, 20)`: 4.2

s. Extension fields: FHJRV 10.28 s per decomposition on the F_{2^155} Oakley curve; Joux–Vitse 8.4 s on F_{p^5} at 32-bit p. Measured relation probability at n = 24 (Trimoska Table 7.7) runs about 2× the 2^{mℓ}/(m!·2^n) heuristic, which the repository's exact census did not reproduce (pdp_followups_20261010 item 6).
- **2022–2026**: no new elliptic-curve factor base. Only Hu 2024 (a critique) and quantum-annealing reruns of Semaev's interval base at 8 and 16 bits.

---

6. Recommended order

1. **Δ-sweep** at m = 3, ℓ = 7 over n = 21…29 and m = 4, ℓ = 5 (#20), with the existing closure instrument. Hours. It decides whether the degree laws are laws in ℓ or artefacts of Δ ≈ 0, which changes how every other cell is read.
2. **Closure degrees at m = 5, 6** on the chain at N ≤ 56 (#10), same instrument, same session. The first points anyone would have on the m ≥ 5 half of the arity curve at the program's own degree measure; compare with HPST15's F4 step degrees 4.3–5.0.
3. **Build GG14 Experiment 2** (#16) and run it beside the chain at m = 4, ℓ = 3, 4. Two to three days. It is the published best m = 4 formulation and the repository has no arm for it.
4. Build the FGHR m = 4 system (#1) and the e_k presentation above ℓ = 6 (#17); measure beside the chain on the same draws. These decide whether "one degree above m = 3" belongs to the arity or to the chain.
5. Run the m = 4, Δ ≥ 0, ℓ ≥ 4 chained cells (#2).
6. Measure w and the rank/density profile on the real degree-4 matrices (#4); re-price the m = 4 table in rho units with the fitted w.
7. The guess-count curve at ℓ = 7, 8 (#3), on a 25–70 GB host.
8. Chain topology (#18) and disjoint bases (#19) as one-day builder additions, read together with items 1–2.
9. The one-afternoon falsifiers that have sat unrun since 2026-10-01/07: G2 at n = 13–19, G4, N1, the cyclotomic solver clause, the explicit trace row (#6–8, #21).
10. The reference arm (#15), because the program's stated goal since 2026-10-22 is to beat published index calculus at matched (q, n, m) and no such baseline exists.

---

7. What this note does not establish

It does not show any factor base or presentation beating rho or published index calculus. The §1.3 what-if is arithmetic on quoted numbers under an assumption (bounded closure degree on a formulation that does not yet exist) that the literature's own heuristics contradict; it is there to show where the arity curve's leverage sits, not to price an attack. Every n = 131 figure is an extrapolation from n ≤ 19.

---

8. Files, build, and graphs checked

- README.md: this note. lit_factor_bases.md, lit_degree_reduction.md, lit_large_arity.md: the three literature lane reports with their data tables and URLs (web-only; eprint PDFs were behind a bot wall and the entries marked abstract-level say so).
- figures/plot.py lays out the recorded and derived values of §1 into figures/degree_vs_ell.svg (repository instruments beside the published F4 step degrees) and figures/arity_curve.svg (the derived arity curve at n = 131), with 150 dpi PNG renders. It computes nothing new; every value is cited in the script header.
- report.typ and report.pdf: built by sh build.sh (Pandoc to Typst, then Typst), with the two figures embedded.
- **Canonical graphs checked and left unchanged**, because no operation count, verdict or coverage moved: docs/index-calculus-scoreboard.html, docs/ic/progress-timeline.json, docs/ic/leaderboard.json, docs/ic/BOUNDARY_TARGETS.md, docs/ic/factor-base-encyclopedia/figures/. The encyclopedia's C8 entry and the degree-reduction note's L2 verdict are the two records this note says were closed on a different experiment than the one the literature measured (§3 rows 16–17); correcting them is a separate change once the proposed cells have run.
- Conductor could not be reached from this session (the control plane at 127.0.0.1:8491 refused the connection under the sandbox), so no task claim was recorded for this path.
- **Lead atlas** (crypto-autoresearcher/knowledge/leads/ecdlp/, reviewed 2026-10-24, 147 leads; matched with tools/lead_atlas.py --match after the note was written). The rows of §3 map onto it as follows, so that they advance a lead's next_action rather than file duplicates. *Advance:* LEAD-ICE-descent-closure-degree-at-scale (open, blocked on a ≥ 250 GB host for the (3, 19, 7) degree-5 cell) gains the Δ-sweep (#20) and the m = 5, 6 cells (#10), both inside the current engine and host class; LEAD-ICE-chained-semaev-dreg-vs-semiregular and LEAD-ICE-linearization-excess-fb-prior gain the vocabulary mapping of §5.2; LEAD-ICE-hybrid-guess-determine-l7 is #3; LEAD-ICE-semaev-yield-model-coset-typing (coset-typed bases, open) is #19 with the published GG14 Exp. 3 and Diem II data attached; LEAD-ICB-tau-horner-chain is #6. *Reopen with a stated reason:* LEAD-EXO-curve-model-representation is closed on the finding that the model-native solving degree is invariant across Weierstrass, Edwards and other models at m = 2; #16 is not a model change but the tuple-level (Z/4)^{m−1} ⊡ S_m invariants GG14 measured at m = 4, which that closure did not test. LEAD-ICB-decomposition-encoding-and-sat is closed on the measured per-attempt slope 2^{αℓ} of SAT and enumeration encodings; #17 concerns the Gröbner cost of the symmetric presentation at fixed ℓ = 6–8, which HPST 2013 and GG14 measured and the closure's slope data do not cover. LEAD-ICE-semaev-chain-cost-model is closed under Semaev's cost model (t* = m* at n = 250–600); #18 asks for the closure *degree* against the auxiliary count, which that model assumes rather than measures. Related known results: KR-IC-fbdb61 (last fall and solving degree), KR-IC-955fd6 (symmetries and torsion), KR-IC-8b9daa (Karabina), KR-IC-f5c584 (trace morphism).

Figures

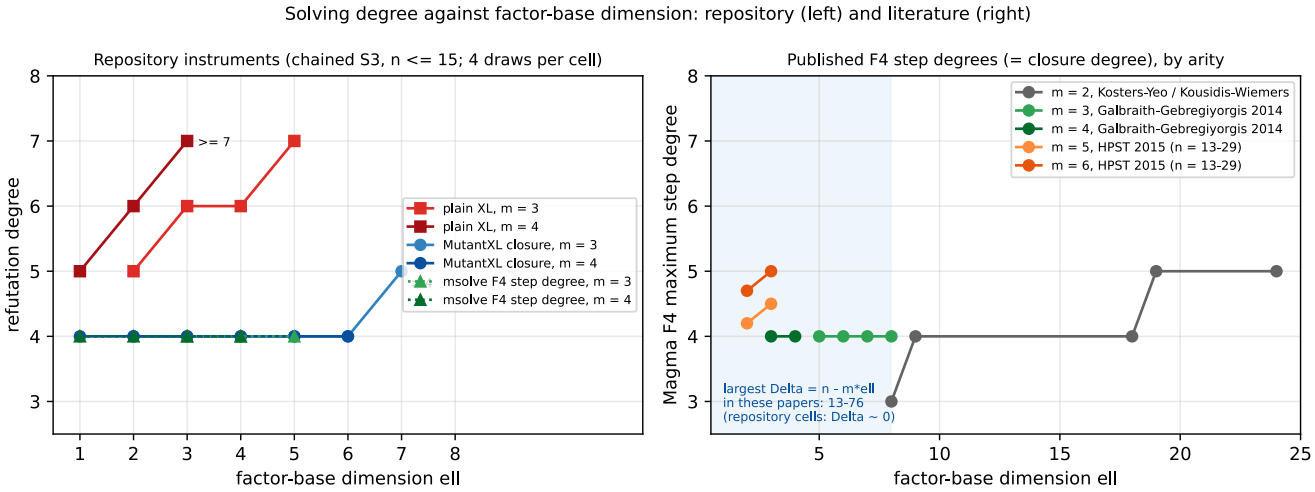


Figure 1: Refutation degree against factor-base dimension. Left: the repository's three instruments on the chained S3 system (plain XL, MutantXL closure, msolve F4), n <= 15, four draws per cell; hollow labels mark lower bounds. Right: published Magma F4 maximum step degrees, the same quantity as the closure, all measured at overdetermination Delta = n - m ell of 13 to 76. Sources in figures/plot.py.

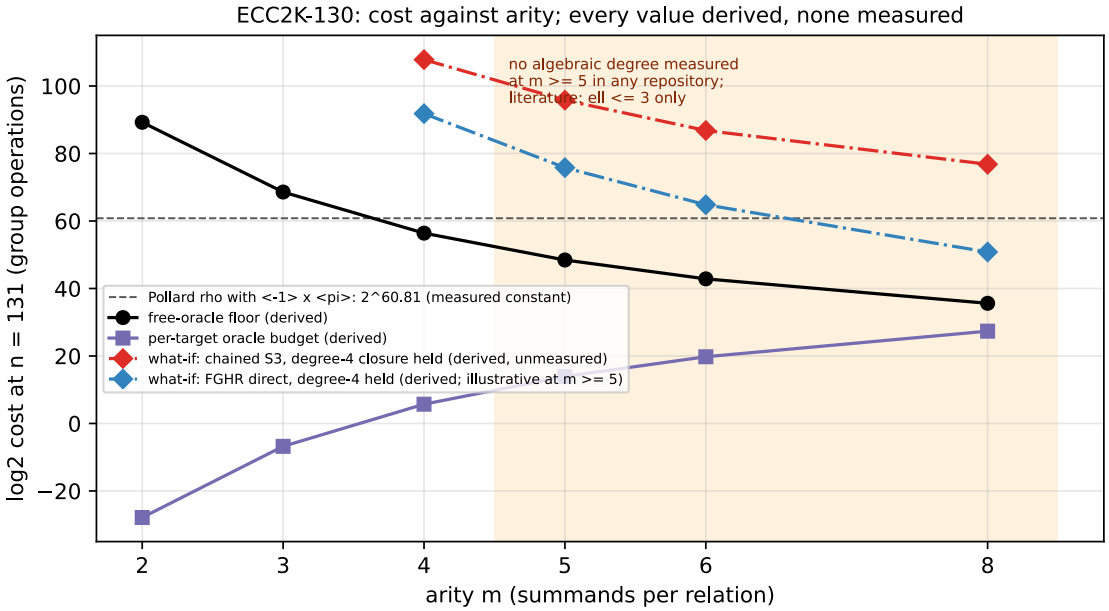


Figure 2: Cost against arity at n = 131 (ECC2K-130). Free-oracle floor and per-target budget from RESEARCH_ECC2K130_DECOMPOSITION.md section 5.3; the what-if rows hold a degree-4 closure, w = 2.2 and an 8-bit column-touch to rho-step conversion, and are derived, not measured (section 1.3). The shaded region has no algebraic degree measurement in any repository.